The IBP Formula

Integration by parts may be written as

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

In the tabular method, the left column is differentiated repeatedly, the right column is integrated repeatedly, and the answer is read from the diagonal products with alternating signs.

The Tabular Method

Choose

$$u = \text{the factor to differentiate}, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \text{the factor to integrate}$$

Then:

1. write the alternating signs $+, -, +, -, \ldots$,
2. differentiate the u-column downward,
3. integrate the $\dfrac{\mathrm{d}v}{\mathrm{d}x}$-column downward,
4. read each arrow diagonally to form the terms,
5. and include the final remainder integral when needed (if the Differentiate column didn't reach 0) by adding the integral of the product of the final row.

Example 1: $\int x^2 e^x \,\mathrm{d}x$

Take

$$u = x^2, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = e^x$$

Sign	Differentiate	Integrate
$+$	x^2	e^x
$-$	$2x$	e^x
$+$	2	e^x
$-$	0	e^x

So

$$\int x^2 e^x \,\mathrm{d}x = x^2 e^x - 2xe^x + 2e^x$$

Hence

$$\boxed{\int x^2 e^x \,\mathrm{d}x = e^x(x^2 - 2x + 2) + C}$$

Example 2: $\int x^3 \sin x \, \mathrm{d}x$

Take

$$u = x^3, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$$

Sign	Differentiate	Integrate
+	x^3	$\sin x$
−	$3x^2$	$-\cos x$
+	$6x$	$-\sin x$
−	6	$\cos x$
+	0	$\sin x$

Reading the diagonals gives

$$\boxed{\int x^3 \sin x \, \mathrm{d}x = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C}$$

Example 3: $\int x^4 e^{2x} \, \mathrm{d}x$

Take

$$u = x^4, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = e^{2x}$$

Sign	Differentiate	Integrate
+	x^4	e^{2x}
−	$4x^3$	$\frac{1}{2}e^{2x}$
+	$12x^2$	$\frac{1}{4}e^{2x}$
−	$24x$	$\frac{1}{8}e^{2x}$
+	24	$\frac{1}{16}e^{2x}$
−	0	$\frac{1}{32}e^{2x}$

So

$$\int x^4 e^{2x} \, \mathrm{d}x = \frac{1}{2}x^4 e^{2x} - x^3 e^{2x} + \frac{3}{2}x^2 e^{2x} - \frac{3}{2}x e^{2x} + \frac{3}{4}e^{2x} + C$$

Hence

$$\boxed{\int x^4 e^{2x} \, \mathrm{d}x = e^{2x}\left(\frac{1}{2}x^4 - x^3 + \frac{3}{2}x^2 - \frac{3}{2}x + \frac{3}{4}\right) + C}$$

Example 4: $\int x \ln x \, \mathrm{d}x$

Choose the logarithm for the differentiated column:

$$u = \ln x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$$

Sign	Differentiate	Integrate
+	$\ln x$	x
−	$\frac{1}{x}$	$\frac{x^2}{2}$

From the first diagonal term,

$$\int x \ln x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, \mathrm{d}x$$

Therefore

$$\int x \ln x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

So

$$\boxed{\int x \ln x \, \mathrm{d}x = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C}$$

Example 5: $\int e^x \sin x \, \mathrm{d}x$

Take

$$u = \sin x, \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = e^x$$

Sign	Differentiate	Integrate
+	$\sin x$	e^x
−	$\cos x$	e^x
+	$-\sin x$	e^x

Let

$$I = \int e^x \sin x \, \mathrm{d}x$$

The table gives

$$I = e^x \sin x - e^x \cos x + \int (-\sin x) e^x \, \mathrm{d}x$$

Since

$$\int (-\sin x) e^x \, \mathrm{d}x = -I$$

we obtain

$$I = e^x \sin x - e^x \cos x - I$$

Hence

$$2I = e^x (\sin x - \cos x) \qquad I = \frac{1}{2} e^x (\sin x - \cos x) + C$$

So

$$\boxed{\int e^x \sin x \, \mathrm{d}x = \frac{1}{2} e^x (\sin x - \cos x) + C}$$